The due date for submitting this assignment has passed.
Due on 2026-03-30, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-03-20, 08:51 IST

In this activity, we will take a look at the application of geogebra to visualize linear transformations $T : \mathrm{R}^2 \rightarrow \mathrm{R}^2$. The notation in the questions that follow will be as in the following project

https://www.geogebra.org/m/krqstvdm

The description of the elements in the project is as follows:

• The parameters a, b, c and d are variables used to determine the linear transformation whose matrix representation with respect to the standard bases is $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

• $Te1$ and $Te2$ are vectors which represent the images of the standard basis vectors $e1$ and $e2$, respectively.

• u is an arbitrary vector in the plane, and Tu is its image under T.

• P is a quadrilateral in the plane and TP is its image under T.

• S is a circular sector in the plane and TS represents its image under T.

To specify the transformation, one can specify the values of a, b, c and d. Or alternatively, move the vectors $Te1$ and $Te2$ to specify the image of the standard basis, thus determining the linear transformation. Once a linear transformation is fixed, the vector u can be moved by dragging its arrowhead. The vector Tu changes accordingly to give the image of the vector u. Similarly, the elements P and S can be moved around and TP and TS can be analyzed to understand the geometric properties of the linear transformation. Answer the following questions using the given project.

1) Consider the linear transformation T which scales each of the standard basis vectors by a non-zero constant. In the geogebra project, this corresponds to moving the vectors T e1 and T e2 so that the arrowheads are on $(\lambda, 0)$ and $(0, \lambda)$, respectively, for some $\lambda \in \mathrm{R} \setminus \{0\}$ of your choice. Which of the following statements are true for such a linear transformation? ***1 point***

- [x] The matrix of T with respect to the standard bases for the domain and codomain is a scalar matrix.
- [x] If λ < 0, the angle between u and T(u) is 180° , for all u ∈ R^2 \ {(0, 0)}.
- [ ] If P is the square with vertices (0, 0),(1, 0),(1, 1) and (0, 1), then its image T P has area equal to λ square units.
- [x] The linear transformation T is always a bijection.
- [ ] If λ > 0, then there is a non-zero vector u such that T(u) = u.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The matrix of T with respect to the standard bases for the domain and codomain is a scalar matrix.
If λ < 0, the angle between u and T(u) is 180° , for all u ∈ R^2 \ {(0, 0)}.
The linear transformation T is always a bijection.

2) It is known that the matrix of a linear transformation T which rotates every vector counter-clockwise by an angle of θ is given by

$$\begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix}$$

In the geogebra project, set

a = 0.6, b = −0.8, c = 0.8, d = 0.6

This gives a transformation which rotates every vector in the plane. Consider the sector S whose vertices are (0, 0),(1, 2) and (2, 1). Under the above transformation, the image of S is a sector TS, which is rotated counter-clockwise by an angle $sin^{-1}(0.8)$. If the vertices of TS are (0, 0),(−1, 2) and (0.4, k), find the value of k.

2.2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2.2

1 point

3) Move the head of the vector T e1 to (1, −1). Observe the image T P of the polygon P as you vary the vector T e2 such that its head lies on the line x = −2. There is exactly one point $(-2, y_0)$ such that T P is a line segment. For all other points, the image of P is a quadrilateral. Find the value of y_0.

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

1 point

4) Consider the transformation in the previous question for which the image of P is a line segment. Which of the following statements are true about the transformation? ***1 point***

- [x] The image of the transformation is a line through the origin.
- [ ] The set {(1, 1)} is a basis for the Kernel of the transformation.
- [ ] The matrix of the transformation is invertible.
- [x] There is exactly one line through the origin whose image under the transformation is the singleton set {(0, 0)}.
- [x] The set {(−1, 1)} is a basis for the image of the transformation.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The image of the transformation is a line through the origin.
There is exactly one line through the origin whose image under the transformation is the singleton set {(0, 0)}.
The set {(−1, 1)} is a basis for the image of the transformation.